Sharp minimax rates for spectral density estimation under sequential local differential privacy

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Abstract

We determine the minimax integrated squared error for estimating the spectral density of a centered stationary Gaussian time series when each observation is released under local differential privacy and the release may depend on all earlier private outputs. On a variance-bounded class whose normalized covariances have Sobolev smoothness $s > 1/2$, the risk is, up to constants, $1 \wedge (n\alpha^4)^{-2s/(2s+1)}$, uniformly for $n \geq 2$ and $0 < \alpha \leq 1$. Thus sequential interaction preserves the sharp non-interactive rate, without a logarithmic loss. The lower bound follows by representing a covariance perturbation through independent Gaussian innovations. An innovation that enters only one past raw observation remains protected by the same privacy inequality after the entire past output has been released. Gaussian smoothing and the current private channel then give a divergence bound of order $n\alpha^4$. The matching upper bound is the non-interactive estimator of Issartel and Roueff.

1 The problem and the minimax rate

A spectral density describes the covariance of a stationary time series. Estimating it from locally private measurements is therefore a basic problem in inference on dependence: each data holder transmits a private version of one raw observation, while the statistician sees only the transmitted outputs. Sequential interaction lets a holder consult previous private outputs before choosing the next release.

Issartel and Roueff [2, Theorems 1 and 3] obtain a sharp, log-free minimax rate for Gaussian spectral density estimation when the releases are non-interactive. Their lower bound concerns channels that depend only on their own raw coordinate, and their Section 3.4 leaves the sharp interactive rate open. Butucea, Klockmann and Krivobokova [1, Section 2.4] construct sequential estimators; the global minimax question is stated on p. 5 of their paper. The difference between these models matters for a lower bound. With interaction, the conditional law of the current raw observation depends on the entire private history, whose distribution changes under the competing spectral densities.

We show that this additional dependence does not improve the joint minimax rate on the class below. The argument controls a Gaussian innovation through the full private history, rather than assuming that the privatized coordinates are conditionally independent. It applies when the privacy parameter is allowed to shrink with the sample size. The access rule remains essential: each release may use its own raw observation and previous private outputs, but cannot read earlier raw observations.

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Fix known parameters $s > 1/2$, $C_0 > 1$ and $C_1 > 0$. For an even, nonnegative, 2π-periodic spectral density f, put

$$\gamma_f(h) = \int_{-\pi}^{\pi} f(\lambda) e^{ih\lambda}\, d\lambda, \qquad \rho_f(h) = \frac{\gamma_f(h)}{\gamma_f(0)}, \qquad h \in \mathbb{Z}. \tag{1}$$

The parameter class is

$$\mathcal{F}(C_0, C_1, s) = \left\{ f :\ C_0^{-1} \le \gamma_f(0) \le C_0, \quad \sum_{h \neq 0} |h|^{2s} |\rho_f(h)|^2 \le C_1^2 \right\}. \tag{2}$$

Here and below f is integrable, real and even, and functions are identified almost everywhere. We use Lebesgue measure for the $L^2([-\pi, \pi])$ norm. The class and normalization in (1)–(2) are exactly those in [2, (3)].

The raw data $X_1, \dots, X_n$ are consecutive observations of a centered, real stationary Gaussian process with spectral density f. A sequential mechanism has standard Borel output spaces E_i and Markov kernels $Q_i(dz_i \mid x_i, z_{<i})$, where $z_{<i} = (z_1, \dots, z_{i-1})$. For every history, raw input pair and measurable output event, require

$$Q_i(B \mid x, z_{<i}) \le e^{\alpha} Q_i(B \mid x', z_{<i}). \tag{3}$$

For a fixed raw vector, its transcript law is the chronological iterated product of these kernels. Write $\mathcal{Q}_{n,\alpha}^{\mathrm{seq}}$ for this class, with the own-observation access rule specified above. Define

$$\mathcal{R}_{n,\alpha}^{\mathrm{seq}} = \inf_{Q \in \mathcal{Q}_{n,\alpha}^{\mathrm{seq}}} \inf_{\widehat{f}} \sup_{f \in \mathcal{F}(C_0, C_1, s)} \mathbb{E}_{f,Q} \left\| \widehat{f}(Z_{1:n}) - f \right\|_2^2, \tag{4}$$

where the second infimum is over measurable $L^2([-\pi, \pi])$-valued estimators. The mechanism and estimator may use the known class parameters.

Theorem 1. *For every fixed $s > 1/2$, $C_0 > 1$ and $C_1 > 0$, there are constants $0 < c \le C < \infty$, depending only on these three parameters, such that for every integer $n \ge 2$ and every $0 < \alpha \le 1$,*

$$c \left[1 \wedge (n\alpha^4)^{-2s/(2s+1)} \right] \le \mathcal{R}_{n,\alpha}^{\mathrm{seq}} \le C \left[1 \wedge (n\alpha^4)^{-2s/(2s+1)} \right]. \tag{5}$$

Consequently, along any sequence of admissible parameter pairs, the minimax risk tends to zero if and only if $n\alpha^4$ tends to infinity.

The upper bound is inherited from [2, Theorem 3]. The new argument is a lower bound for every mechanism in (4), including arbitrary dependence on the private history. On a moving-average pair, the innovation shared by the current and a past raw observation occurs only once in the past. Privacy bounds its posterior density relative to its original Gaussian law by $e^{\pm\alpha}$. Adding the fresh innovation makes the two current predictive laws close, and privatizing the current coordinate contracts their divergence again. Each of these two effects contributes a factor of order α^2. A common Gaussian background permits the same calculation for the Fourier alternatives used in the minimax lower bound.

The proof concerns the precise Gaussian class and access rule above. It makes no claim for releases that can read past raw data or for non-Gaussian time series. Section 2 proves the privacy and smoothing facts, Section 3 gives the sequential divergence estimate, and Section 4 proves Theorem 1.

Remark 2. Large-sample upper bounds with an α-dependent threshold do not by themselves give a bound uniform in a shrinking privacy parameter. For example, the clipping calculation in [1, Appendix A.3, p. 27] contains the remainder $\exp[-\widetilde{\tau}_n \alpha/(4\tau_n^2)]$ with $\widetilde{\tau}_n = 16(\log n)^{1+\eta}\tau_n^2$, for fixed $\eta > 0$. It equals $\exp[-4\alpha(\log n)^{1+\eta}]$; its polynomial decay therefore requires an α-dependent large-n condition. This qualification is relevant when comparing their Theorem 7 with (5).

2 Privacy of a Gaussian innovation

We first record how a single raw coordinate can affect the entire output history. This observation uses the restriction on access to past raw data.

Lemma 3. *Let $L_t(\cdot \mid x_{1:t})$ be the law of the first t outputs of a mechanism satisfying (3), with a deterministic raw prefix. If $x_{1:t}$ and $x'_{1:t}$ differ in only one coordinate, then*

$$L_t(B \mid x_{1:t}) \le e^{\alpha} L_t(B \mid x'_{1:t})$$

for every measurable prefix event B. If a latent variable U is independent of a nuisance vector V and the raw prefix $x(U,V)$ depends on U in at most one coordinate, the channel from U to the whole private prefix is also α-LDP.

Proof. Suppose the changed coordinate is j. The histories before j have the same law. At time j, integrate (3) against this common history law to dominate the joint output measure through time j by e^{α} times the other measure. At each later time k, both measures are extended by the same kernel $z_{<k} \mapsto Q_k(\cdot \mid x_k, z_{<k})$, because $x_k = x'_k$. The integral of any nonnegative measurable function preserves a measure domination, so induction gives the claim through time t. For the latent-variable assertion, first fix $V = v$ and apply this comparison to any two values of U. Integrate over the same law of V, which is independent of U. If the coordinate never appears, the channel is constant. □

Lemma 4. *Let K be an α-LDP kernel between standard Borel spaces, with input law π and output law $M = K\pi$. For M-almost every output z, the posterior input law has a density $r(\cdot, z)$ relative to π satisfying*

$$e^{-\alpha} \le r(u,z) \le e^{\alpha} \quad \text{for } \pi\text{-almost every } u, \qquad \int (r(u,z)-1)^2 \pi(du) \le (e^{\alpha}-1)^2. \tag{6}$$

Proof. Averaging the privacy inequality over a second input gives, for every u, $e^{-\alpha} M \le K(\cdot \mid u) \le e^{\alpha} M$ as inequalities of measures. Applying these inequalities to the measurable sections of a product event and integrating in u yields

$$e^{-\alpha}(\pi \otimes M) \le \pi \otimes K \le e^{\alpha}(\pi \otimes M).$$

The Radon–Nikodym derivative $r(u,z)$ of the middle measure has the same bounds. By Fubini and equality of the output marginals, $\int r(u,z)\pi(du) = 1$ for M-almost every z; the measures $r(u,z)\pi(du)$ give a version of the posterior. These uses of Radon–Nikodym and product integration are covered, for example, by [4, Theorems 1.6.6 and 1.6.12]. Finally $|r-1| \le e^{\alpha} - 1$ proves the squared-density bound. □

We write $D(P\|R)$ for relative entropy and $\chi^2(P\|R) = \int (dP/dR - 1)^2 dR$, with value $+\infty$ if the required absolute continuity fails. The next bound measures the effect of adding an independent Gaussian innovation to the posterior in Lemma 4.

Lemma 5. *Let $G = N(0,1)$, and let U have density r relative to G, where $r - 1 \in L^2(G)$. Let $W \sim G$ be independent of U. If $|t| < 1$ and P_t is the law of $tU + \sqrt{1-t^2}W$, then*

$$\chi^2(P_t\|G) \le \frac{t^2}{1-t^2} \int (r-1)^2 dG. \tag{7}$$

If $e^{-\alpha} \le r \le e^{\alpha}$ almost everywhere, the same bounds hold for dP_t/dG.

Proof. Under the reference law $U \sim G$, the pair (U, Y), where $Y = tU + \sqrt{1-t^2}W$, is a bivariate standard normal with covariance

$$\Sigma = \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

Let $S_t(u, y)$ be its density relative to $G \otimes G$. Its two marginals are G, so the density $p_t = dP_t/dG$ satisfies

$$p_t(y) - 1 = \int (r(u) - 1)(S_t(u, y) - 1)G(du).$$

Cauchy–Schwarz and Tonelli give

$$\int (p_t - 1)^2 dG \le \left[\int (r-1)^2 dG\right] \iint (S_t - 1)^2 \, dG \, dG.$$

For $x = (u, y)^{\mathsf{T}}$ the normal density formula gives

$$S_t(x) = (\det \Sigma)^{-1/2} \exp\left[-\tfrac{1}{2} x^{\mathsf{T}} (\Sigma^{-1} - I)x\right].$$

The matrix $2\Sigma^{-1} - I$ has positive eigenvalues $(1-t)/(1+t)$ and $(1+t)/(1-t)$, hence determinant one. The change of variables $(u, y) \mapsto ((u+y)/\sqrt{2}, (u-y)/\sqrt{2})$ evaluates the convergent Gaussian integral as

$$\iint S_t^2 \, dG \, dG = \frac{1}{\det \Sigma \sqrt{\det(2\Sigma^{-1} - I)}} = \frac{1}{1-t^2}.$$

Since $\iint S_t \, dG \, dG = 1$, the integral of $(S_t - 1)^2$ equals $t^2/(1-t^2)$. This proves (7), including $t = 0$. Finally, $p_t(y) = \int r(u) S_t(u, y) G(du)$ is an average of r under the reference conditional normal law, so it retains its upper and lower bounds. □

3 Sequential testing for a covariance perturbation

The testing pair below changes one autocovariance and leaves the marginal variance fixed. A moving-average representation makes the relevant past innovation fall within Lemma 3.

Proposition 6. *For $a > 0$, an integer $h \ge 1$ and $|\delta| \le 1$, define*

$$g_\pm(\lambda) = \frac{a}{2\pi}\left[1 \pm \frac{\delta}{2}\cos(h\lambda)\right].$$

Let $M_\pm$ be the output laws of any sequential mechanism satisfying (3) *under these two centered Gaussian observation laws. For every $n \ge 1$ and $0 < \alpha \le 1$,*

$$D(M_+ \| M_-) \le 128(n-h)_+ \alpha^4 \delta^2, \qquad (n-h)_+ = \max(n-h, 0). \tag{8}$$

Proof. Choose $A > 0$ and signed B by

$$B^2 = \frac{1 - \sqrt{1 - \delta^2/4}}{2}, \qquad A = \sqrt{1 - B^2}, \qquad B = \frac{\delta}{4A}. \tag{9}$$

These relations are consistent and give $A^2 + B^2 = 1$, $AB = \delta/4$, $B^2 \le \delta^2/8$, and $A^2 \ge 7/8$. Take independent standard normals ε_k, $1 - h \le k \le n$, and put

$$U_i^{\pm} = \sqrt{a}\,(A\varepsilon_i \pm B\varepsilon_{i-h}), \qquad 1 \le i \le n. \tag{10}$$

The variance is a, the lag-h covariance is $\pm aAB = \pm a\delta/4$, and all other nonzero-lag covariances vanish. These are exactly the Fourier coefficients of $g_\pm$ in (1). Centered Gaussian vectors are determined by their covariances [3, Section 3.10], so (10) realizes the observation laws.

Both raw prefix laws are mutually absolutely continuous. Indeed $g_\pm \geq a/(4\pi)$, and for every real coefficient vector c their covariance matrices satisfy

$$c^{\mathsf{T}} T_\pm c = \int_{-\pi}^{\pi} g_\pm(\lambda) \left| \sum_j c_j e^{ij\lambda} \right|^2 d\lambda \geq \frac{a}{2} \sum_j c_j^2.$$

They therefore have strictly positive Gaussian densities. Passing them through the same transcript kernel preserves absolute continuity in both directions. In particular, the two laws of each private history have the same null sets. All conditional comparisons that follow are made outside these common null sets; regular conditional laws exist on the standard Borel spaces involved.

Fix time i. The fresh ε_i is independent of the pair $(\varepsilon_{i-h}, Z_{<i})$: no raw prefix coordinate involves ε_i, and the prefix kernels use only that prefix. If $i > h$, the innovation ε_{i-h} appears in the raw prefix only in $U^\pm_{i-h}$. Its next appearance is in the current $U^\pm_i$. Lemma 3, with all other innovations as independent nuisance variables, says that its channel to the entire history is α-LDP. Lemma 4 bounds its posterior density relative to G by $e^{\pm\alpha}$ and its χ^2 divergence by $(e^\alpha - 1)^2$, separately under each sign model.

For a fixed common history z, let $P^z_\pm$ be the predictive law of $U^\pm_i$. Write $G_a = N(0, a)$. Apply Lemma 5 with $t = \pm B$; scaling by $\sqrt{a}$ cancels from both density ratios. Since $B^2/(1 - B^2) \leq \delta^2/(8 - \delta^2) \leq \delta^2/7$, we obtain

$$e^{-\alpha} \leq \frac{dP^z_\pm}{dG_a} \leq e^\alpha, \qquad \chi^2(P^z_\pm \| G_a) \leq \frac{\delta^2}{7}(e^\alpha - 1)^2. \tag{11}$$

When $i \leq h$, ε_{i-h} is absent from the observed prefix. It remains a standard normal independent of the history and of the fresh innovation, so in this case $P^z_+ = P^z_- = G_a$.

For $i > h$, put $p_\pm = dP^z_\pm/dG_a$. Using $\log u \leq u - 1$ and $p_- \geq e^{-\alpha}$ gives

$$\begin{aligned} D(P^z_+ \| P^z_-) &\leq \chi^2(P^z_+ \| P^z_-) \leq e^\alpha \int (p_+ - p_-)^2 dG_a \\ &\leq 2e^\alpha \{\chi^2(P^z_+ \| G_a) + \chi^2(P^z_- \| G_a)\} \leq \frac{4e^\alpha}{7}\delta^2(e^\alpha - 1)^2. \end{aligned} \tag{12}$$

At the same history z, the current kernel $Q_i(\cdot \mid \cdot, z)$ is one common α-LDP channel under both models. By [2, Lemma 3], for any input laws P, R it satisfies

$$D(Q_i P \| Q_i R) \leq \beta_\alpha D(P \| R), \qquad \beta_\alpha = \frac{1}{2}(e^\alpha - e^{-\alpha})^2 \wedge 1. \tag{13}$$

For $0 < \alpha \leq 1$, convexity gives $e^\alpha - 1 \leq (e - 1)\alpha < 2\alpha$, and integration of $e^t + e^{-t} \leq 2e < 6$ gives $e^\alpha - e^{-\alpha} \leq 6\alpha$. Thus

$$\beta_\alpha \frac{4e^\alpha}{7}(e^\alpha - 1)^2 \leq \frac{864}{7}\alpha^4 < 128\alpha^4.$$

The conditional output divergence at each $i > h$ is at most $128\alpha^4\delta^2$, and it is zero at each $i \leq h$. The kernel chain rule [2, Lemma 14] now gives

$$D(M_+ \| M_-) = \sum_{i=1}^{n} \mathbb{E}_+ \left[D(\mathcal{L}_+(Z_i \mid Z_{<i}) \,\|\, \mathcal{L}_-(Z_i \mid Z_{<i})) \right],$$

which proves (8). Finite products of the output spaces are standard Borel, so the chain rule and conditional versions apply throughout. □

Theorem 7. *Let f_+, f_- be even, nonnegative, integrable spectral densities. Suppose that for some $a > 0$, integer $h \geq 1$ and $|\delta| \leq 1$,*

$$\min(f_+, f_-) \geq \frac{3a}{4\pi}, \qquad f_+ - f_- = \frac{a\delta}{2\pi}\cos(h\lambda) \quad \text{almost everywhere.} \tag{14}$$

Every sequential α-LDP mechanism with the specified access rule satisfies

$$D(\mathcal{L}_{f_+,Q}(Z_{1:n}) \,\|\, \mathcal{L}_{f_-,Q}(Z_{1:n})) \leq 128(n-h)_+\alpha^4\delta^2 \tag{15}$$

for $n \geq 1$ and $0 < \alpha \leq 1$.

Proof. With $g_\pm$ as in Proposition 6, define the common spectral background

$$f_V = \frac{f_+ + f_-}{2} - \frac{a}{2\pi} \geq \frac{a}{4\pi}.$$

Let $U^\pm$ be the centered Gaussian vectors with spectral densities $g_\pm$ constructed in (10), and let $M_\pm$ denote the transcript laws under $f_\pm$ and Q. The background f_V has a positive definite finite Toeplitz covariance matrix, by the same quadratic-form calculation used above. A Cholesky factor therefore constructs a centered Gaussian vector V with that covariance, independent of $U^\pm$. The vectors $X^\pm = U^\pm + V$ have the original covariance matrices because $g_\pm + f_V = f_\pm$. Their centered Gaussian laws are the required raw laws.

Condition on $V = v$. The release kernels become

$$Q_i^v(dz_i \mid u_i, z_{<i}) = Q_i(dz_i \mid u_i + v_i, z_{<i}).$$

For every fixed v, these are still sequential α-LDP kernels: the raw shift preserves (3), and each kernel still reads only its current u_i and previous private outputs. Proposition 6 therefore applies uniformly in v. The law ν of V is common to both models. The common-mixture divergence inequality [2, Lemma 15], or the joint chain rule followed by projection, gives

$$D(M_+\|M_-) \leq \int D(\mathcal{L}_+(Z_{1:n} \mid V = v) \,\|\, \mathcal{L}_-(Z_{1:n} \mid V = v))\nu(dv).$$

This proves (15). The conditional transcript kernels are measurable in v by iterated kernel composition. □

4 Proof of the minimax bounds

Proof of Theorem 1. For the lower bound, write $b = n\alpha^4$ and choose

$$m = \left\lceil (\max(1, b))^{1/(2s+1)} \right\rceil, \qquad \kappa = \min\left(\frac{C_1}{\sqrt{2}}, \frac{1}{512}\right), \qquad \theta = \kappa m^{-s-1/2}. \tag{16}$$

The choice of m covers both small and large effective sample sizes. The two restrictions on κ enforce the class constraint and keep neighboring output laws close. For $\omega \in \{-1, 1\}^m$, set

$$f_\omega(\lambda) = \frac{1}{2\pi}\left[1 + 2\theta\sum_{h=1}^{m}\omega_h\cos(h\lambda)\right]. \tag{17}$$

Since $s > 1/2$, $\theta m \le \kappa$ and $f_\omega \ge (1-2\kappa)/(2\pi) > 0$. These densities are even and periodic. Direct integration gives $\gamma_\omega(0) = 1$, $\gamma_\omega(\pm h) = \theta\omega_h$ for $1 \le h \le m$, and zero at other nonzero lags. Hence the variance is in $[C_0^{-1}, C_0]$, and

$$\sum_{h\neq 0} |h|^{2s}|\rho_\omega(h)|^2 = 2\theta^2 \sum_{h=1}^{m} h^{2s} \le 2\kappa^2 \le C_1^2.$$

Thus all f_ω are in the original class. Their positive spectral lower bound makes every finite Toeplitz covariance positive definite. The corresponding Gaussian finite laws are consistent, so the countable extension theorem [3, Theorem A.3.1] gives a stationary Gaussian process: translation invariance follows from its covariance and Gaussian identification. This also verifies the observation-law realization of the subfamily.

If ω, ω' differ only at coordinate h, then

$$f_\omega - f_{\omega'} = \frac{2\theta\omega_h}{\pi}\cos(h\lambda).$$

Apply Theorem 7 with $a = 1/4$ and $\delta = 16\theta\omega_h$. Here $|\delta| \le 1/32$, and $\min(f_\omega, f_{\omega'}) \ge 3/(16\pi)$, as required by (14). For any admissible mechanism, let μ_ω be its entire output law under f_ω. Then

$$D(\mu_\omega \| \mu_{\omega'}) \le 32768 b\theta^2 \le 32768\kappa^2 \le \frac{1}{8}, \tag{18}$$

because $m^{2s+1} \ge \max(1, b)$.

For any measurable L^2-valued estimator, define

$$T_h = \mathrm{Re}\int_{-\pi}^{\pi} \widehat{f}(\lambda)\cos(h\lambda)\,d\lambda, \qquad \widehat{\omega}_h = \begin{cases} 1, & T_h \ge 0, \\ -1, & T_h < 0. \end{cases}$$

The coefficient functionals are continuous by Cauchy–Schwarz, so these decisions are measurable. Since $\int f_\omega \cos(h\lambda)d\lambda = \theta\omega_h$ and the cosine functions are orthogonal with squared norm π, their finite orthogonal projection gives

$$\left\|\widehat{f} - f_\omega\right\|_2^2 \ge \frac{1}{\pi}\sum_{h=1}^{m}(T_h - \theta\omega_h)^2 \ge \frac{\theta^2}{\pi} d_H(\widehat{\omega}, \omega). \tag{19}$$

For a complex estimator, first take the real part of its error; this cannot increase its norm. The second inequality includes zero ties, since a wrong sign costs at least θ in its coefficient. Here d_H is the number of differing sign coordinates.

The Assouad–Pinsker inequality [2, Lemma 7] applies to any family of probability laws on a common measurable space. It states

$$\inf_T \sup_\omega \mathbb{E}_\omega d_H(T, \omega) \ge \frac{m}{2} \min_{d_H(\omega,\omega')=1} \left[1 - \sqrt{\frac{D(\mu_\omega\|\mu_{\omega'})}{2}}\right] \ge \frac{3m}{8},$$

where the last bound is (18). Applying it to the actual sequential output family and using (19) yields, for every Q,

$$\inf_{\widehat{f}} \sup_{f\in\mathcal{F}(C_0,C_1,s)} \mathbb{E}_{f,Q}\left\|\widehat{f} - f\right\|_2^2 \ge \frac{3\kappa^2}{8\pi} m^{-2s}.$$

The same inequality holds after taking the infimum over Q. As $m \le 2(\max(1,b))^{1/(2s+1)}$, the lower bound in (5) holds with $c = 3\kappa^2/(8\pi 2^{2s}) > 0$.

For the upper bound, [2, Theorem 3] supplies a non-interactive mechanism and a measurable estimator on exactly (2) with

$$\sup_{f\in\mathcal{F}(C_0,C_1,s)} \mathbb{E}_{f,Q}\left\|\widehat{f}-f\right\|_2^2 \le C_{C_0,C_1,s}\left[1\wedge\{n(\alpha^4\wedge 1)\}^{-2s/(2s+1)}\right]$$

for every $n\ge 2$ and $\alpha>0$. Their release adds independent Laplace noise of scale $3/\alpha$ to the binary indicator $\mathbf{1}_{\{|X_i|>C_0\}}\in\{0,1\}$ and the sign statistic $\operatorname{sgn}(X_i)\in[-1,1]$. The joint ℓ^1 sensitivity is at most $1+2=3$, so this is an α-LDP kernel into $\mathbb{R}^2$. A non-interactive kernel is sequentially admissible by ignoring the private history. For $\alpha\le 1$, the displayed expression is the upper bound in (5).

Finally, the two-sided estimate shows that the risk tends to zero when $n\alpha^4\to\infty$. If $n\alpha^4$ fails to tend to infinity, it has a bounded subsequence; the lower bound is bounded away from zero there. This proves the converse and completes the theorem. □

The effective size $n\alpha^4$ therefore describes both the consistent regime and its boundary. In particular, every regime with bounded $n\alpha^4$ prevents uniformly consistent spectral density estimation, and no logarithmic factor is required in either minimax bound.

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